

```

import sys
print(sys.executable)

d:\B-Tech\7th Sem_BTech\ADS_Lab\Assignment_01\tfenv\Scripts\python.exe

# Import Libraries

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Scikit-learn
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder, StandardScaler
from sklearn.metrics import mean_absolute_error, mean_squared_error,
r2_score

# TensorFlow/Keras
import tensorflow as tf
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense

# Load Dataset
df = pd.read_csv("Housing.csv")

# Display first 5 rows
df.head()

```

	price	area	bedrooms	bathrooms	stories	mainroad	guestroom
0	13300000	7420	4	2	3	yes	no
1	12250000	8960	4	4	4	yes	no
2	12250000	9960	3	2	2	yes	no
3	12215000	7500	4	2	2	yes	no
4	11410000	7420	4	1	2	yes	yes

	hotwaterheating	airconditioning	parking	prefarea	furnishingstatus
0	no	yes	2	yes	furnished
1	no	yes	3	no	furnished
2	no	no	2	yes	semi-furnished
3	no	yes	3	yes	furnished
4	no	yes	2	no	furnished

```

# Shape of dataset
print("Shape :", df.shape)

```

```

# Column names
print("\nColumns:")
print(df.columns)

# Information
print("\nDataset Information:")
df.info()

# Statistical Summary
print("\nStatistical Summary:")
print(df.describe())

```

Shape : (545, 13)

Columns:

```

Index(['price', 'area', 'bedrooms', 'bathrooms', 'stories',
       'mainroad',
       'guestroom', 'basement', 'hotwaterheating', 'airconditioning',
       'parking', 'prefarea', 'furnishingstatus'],
      dtype='str')

```

Dataset Information:

<class 'pandas.DataFrame'>

RangeIndex: 545 entries, 0 to 544

Data columns (total 13 columns):

#	Column	Non-Null Count	Dtype
0	price	545 non-null	int64
1	area	545 non-null	int64
2	bedrooms	545 non-null	int64
3	bathrooms	545 non-null	int64
4	stories	545 non-null	int64
5	mainroad	545 non-null	str
6	guestroom	545 non-null	str
7	basement	545 non-null	str
8	hotwaterheating	545 non-null	str
9	airconditioning	545 non-null	str
10	parking	545 non-null	int64
11	prefarea	545 non-null	str
12	furnishingstatus	545 non-null	str

dtypes: int64(6), str(7)

memory usage: 55.5 KB

Statistical Summary:

	price	area	bedrooms	bathrooms	stories
count	5.450000e+02	545.000000	545.000000	545.000000	545.000000
mean	4.766729e+06	5150.541284	2.965138	1.286239	1.805505

std	1.870440e+06	2170.141023	0.738064	0.502470	0.867492
min	1.750000e+06	1650.000000	1.000000	1.000000	1.000000
25%	3.430000e+06	3600.000000	2.000000	1.000000	1.000000
50%	4.340000e+06	4600.000000	3.000000	1.000000	2.000000
75%	5.740000e+06	6360.000000	3.000000	2.000000	2.000000
max	1.330000e+07	16200.000000	6.000000	4.000000	4.000000

```

    parking
count  545.000000
mean    0.693578
std     0.861586
min     0.000000
25%     0.000000
50%     0.000000
75%     1.000000
max     3.000000

```

```
print(df.dtypes)
```

```

price          int64
area           int64
bedrooms       int64
bathrooms      int64
stories        int64
mainroad       str
guestroom      str
basement       str
hotwaterheating str
airconditioning str
parking        int64
prefarea       str
furnishingstatus str
dtype: object

```

```
plt.figure(figsize=(10,8))
```

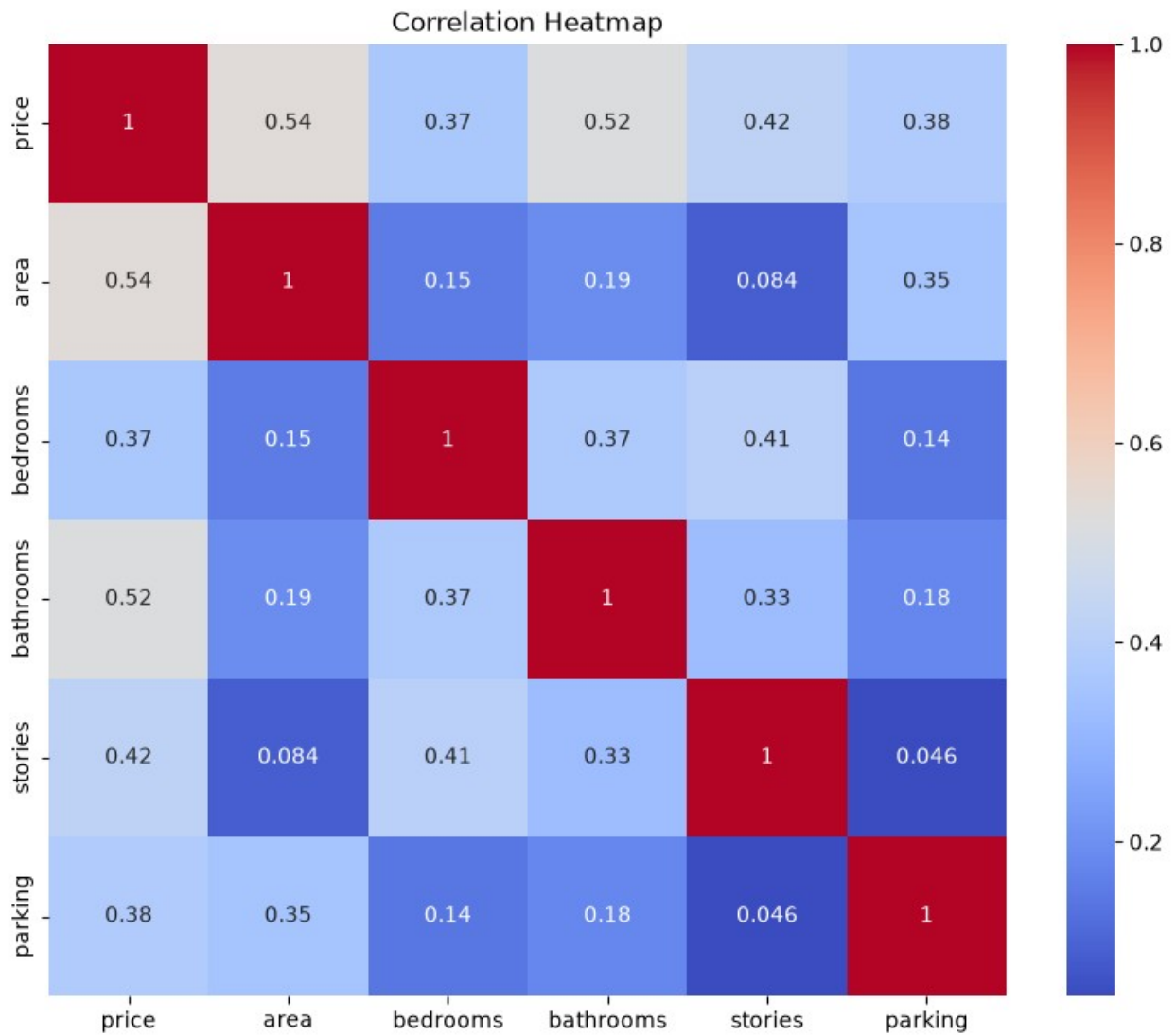
```
numeric_df = df.select_dtypes(include=['int64','float64'])
```

```

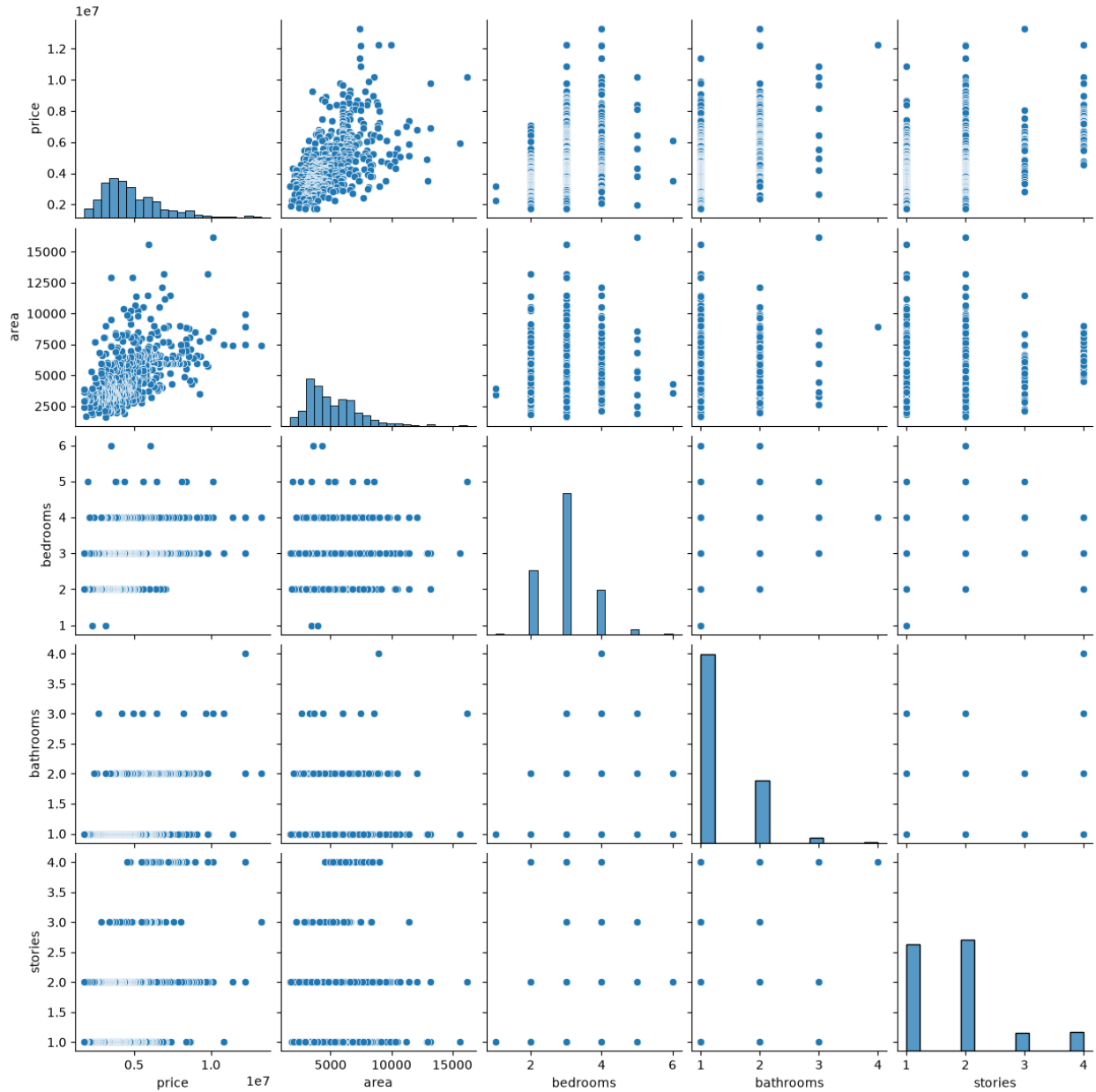
sns.heatmap(
    numeric_df.corr(),
    annot=True,
    cmap='coolwarm'
)

```

```
plt.title("Correlation Heatmap")
plt.show()
```



```
sns.pairplot(df[['price', 'area', 'bedrooms', 'bathrooms', 'stories']])
plt.show()
```

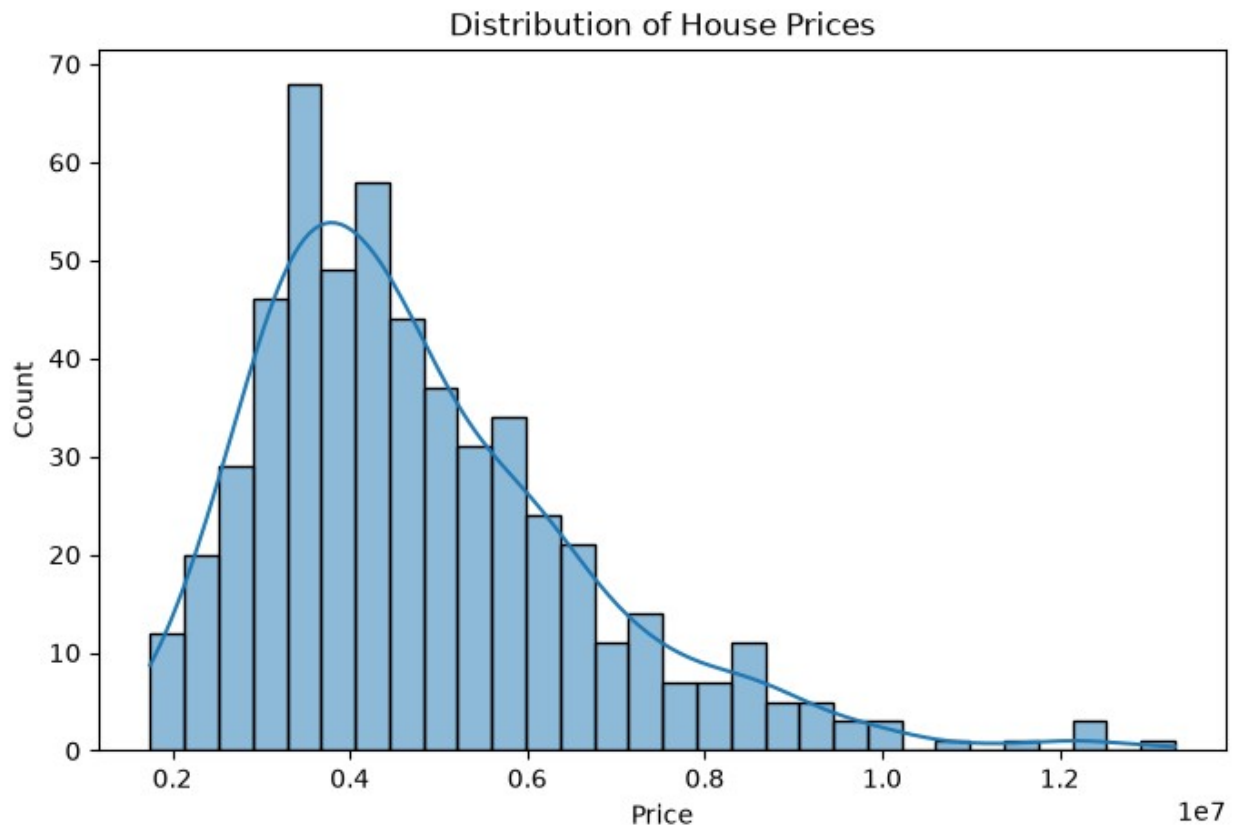


```
plt.figure(figsize=(8,5))

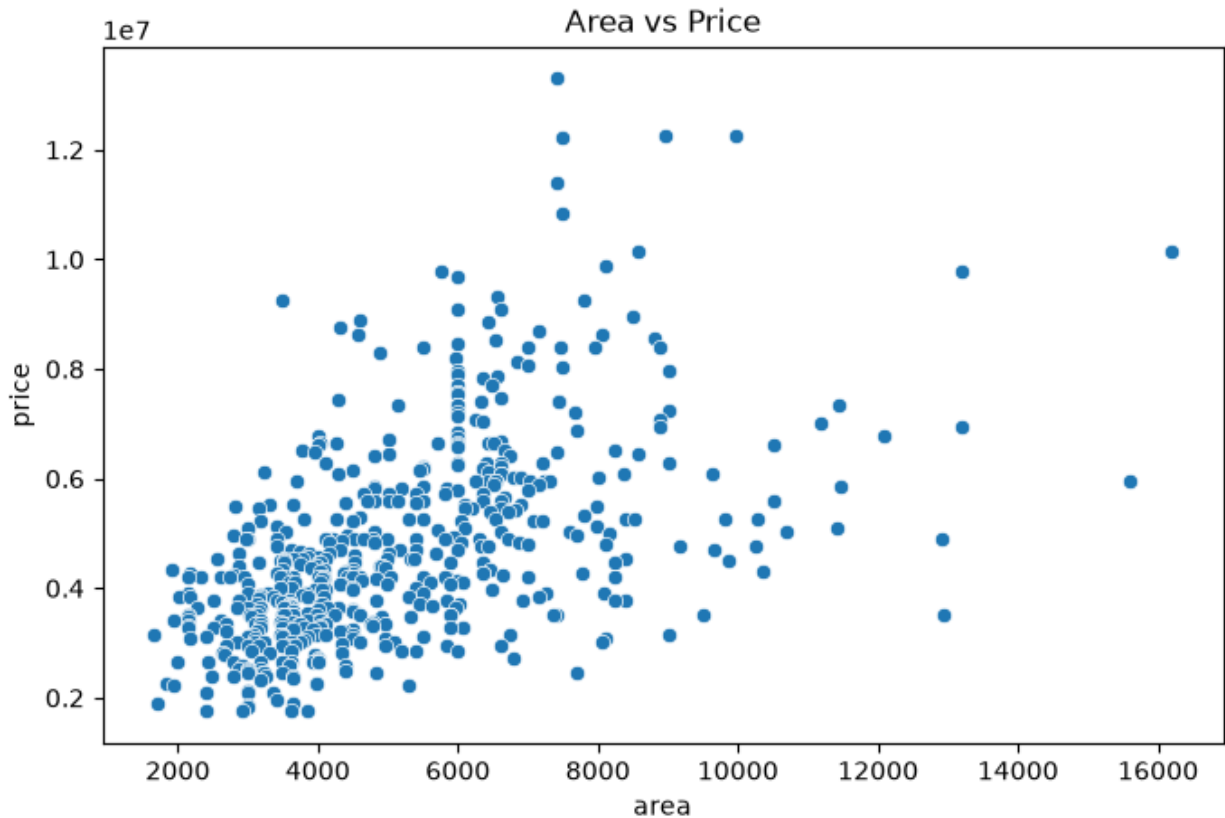
sns.histplot(df['price'], bins=30, kde=True)

plt.title("Distribution of House Prices")
plt.xlabel("Price")
plt.ylabel("Count")

plt.show()
```



```
plt.figure(figsize=(8,5))
sns.scatterplot(x='area', y='price', data=df)
plt.title("Area vs Price")
plt.show()
```



```
# Encode categorical variables
```

```
from sklearn.preprocessing import LabelEncoder
```

```
le = LabelEncoder()
```

```
categorical_columns = df.select_dtypes(include='object').columns
```

```
for col in categorical_columns:
    df[col] = le.fit_transform(df[col])
```

```
# Display first 5 rows after encoding
```

```
df.head()
```

C:\Users\siddh\AppData\Local\Temp\ipykernel_6160\3967287033.py:7:
Pandas4Warning: For backward compatibility, 'str' dtypes are included by select_dtypes when 'object' dtype is specified. This behavior is deprecated and will be removed in a future version. Explicitly pass 'str' to 'include' to select them, or to 'exclude' to remove them and silence this warning.

See https://pandas.pydata.org/docs/user_guide/migration-3-strings.html#string-migration-select-dtypes for details on how to write code that works with pandas 2 and 3.

```
categorical_columns = df.select_dtypes(include='object').columns
```

	price	area	bedrooms	bathrooms	stories	mainroad	
guestroom \							
0	13300000	7420	4	2	3	1	0
1	12250000	8960	4	4	4	1	0
2	12250000	9960	3	2	2	1	0
3	12215000	7500	4	2	2	1	0
4	11410000	7420	4	1	2	1	1

	basement	hotwaterheating	airconditioning	parking	prefarea \
0	0	0	1	2	1
1	0	0	1	3	0
2	1	0	0	2	1
3	1	0	1	3	1
4	1	0	1	2	0

	furnishingstatus
0	0
1	0
2	1
3	0
4	0

Features and Target

```
X = df.drop("price", axis=1)
y = df["price"]
```

```
print("Features Shape :", X.shape)
print("Target Shape :", y.shape)
```

```
Features Shape : (545, 12)
Target Shape : (545,)
```

```
from sklearn.model_selection import train_test_split
```

```
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.20,
    random_state=42
)
```

```
print("Training Data :", X_train.shape)
print("Testing Data :", X_test.shape)
```


Training Data : (436, 12)

Testing Data : (109, 12)

```
from sklearn.preprocessing import StandardScaler
```

```
scaler = StandardScaler()
```

```
X_train = scaler.fit_transform(X_train)
```

```
X_test = scaler.transform(X_test)
```

```
print(X_train[:5])
```

```
[[ 0.38416819  0.05527092  1.53917323  2.58764353  0.40715525 -  
0.46677307
```

```
 -0.74642003 -0.23052136  1.50124327  0.36795665 -0.55262032 -  
1.40940148]
```

```
 [ 0.9291807   0.05527092  1.53917323 -0.91249891  0.40715525 -  
0.46677307
```

```
  1.33972825 -0.23052136  1.50124327  2.70998729 -0.55262032 -  
0.07931697]
```

```
 [-0.60775457 -1.28351359 -0.5579503  -0.91249891  0.40715525 -  
0.46677307
```

```
  1.33972825 -0.23052136  1.50124327  1.53897197 -0.55262032 -  
1.40940148]
```

```
 [-1.15549214  0.05527092 -0.5579503   0.25421524  0.40715525 -  
0.46677307
```

```
  1.33972825 -0.23052136 -0.66611456 -0.80305867  1.80956067  
1.25076755]
```

```
 [-0.63773026  0.05527092 -0.5579503   0.25421524  0.40715525 -  
0.46677307
```

```
 -0.74642003 -0.23052136 -0.66611456 -0.80305867 -0.55262032  
1.25076755]]
```

```
print("X_train :", X_train.shape)
```

```
print("X_test :", X_test.shape)
```

```
print("y_train :", y_train.shape)
```

```
print("y_test :", y_test.shape)
```

```
X_train : (436, 12)
```

```
X_test : (109, 12)
```

```
y_train : (436,)
```

```
y_test : (109,)
```

```
# Import Required Libraries
```

```
from tensorflow.keras.models import Sequential
```

```
from tensorflow.keras.layers import Dense
```

```
# Create ANN Model
```

```
model = Sequential()
```

```
# Input Layer + Hidden Layer 1
model.add(Dense(64, activation='relu',
input_shape=(X_train.shape[1],)))
```

```
# Hidden Layer 2
model.add(Dense(32, activation='relu'))
```

```
# Hidden Layer 3
model.add(Dense(16, activation='relu'))
```

```
# Output Layer (Regression)
model.add(Dense(1, activation='linear'))
```

```
d:\B-Tech\7th Sem_BTech\ADS_Lab\Assignment_01\tfenv\Lib\site-packages\
keras\src\layers\core\dense.py:107: UserWarning: Do not pass an
`input_shape`/`input_dim` argument to a layer. When using Sequential
models, prefer using an `Input(shape)` object as the first layer in
the model instead.
```

```
    super().__init__(activity_regularizer=activity_regularizer,
**kwargs)
```

```
model.summary()
```

```
Model: "sequential"
```

Layer (type) Param #	Output Shape	
dense (Dense) 832	(None, 64)	
dense_1 (Dense) 2,080	(None, 32)	
dense_2 (Dense) 528	(None, 16)	
dense_3 (Dense) 17	(None, 1)	

```
Total params: 3,457 (13.50 KB)
```

```
Trainable params: 3,457 (13.50 KB)
```

Non-trainable params: 0 (0.00 B)

```
model.compile(  
    optimizer='adam',  
    loss='mse',  
    metrics=['mae']  
)
```

WARNING:tensorflow:TensorFlow GPU support is not available on native Windows for TensorFlow >= 2.11. Even if CUDA/cuDNN are installed, GPU will not be used. Please use WSL2 or the TensorFlow-DirectML plugin.

```
history = model.fit(  
    X_train,  
    y_train,  
    epochs=100,  
    batch_size=32,  
    validation_split=0.2,  
    verbose=1  
)
```

Epoch 1/100

11/11 _____ 1s 16ms/step - loss: 25349387714560.0000 -
mae: 4727421.5000 - val_loss: 24781596393472.0000 - val_mae:
4623896.5000

Epoch 2/100

11/11 _____ 0s 6ms/step - loss: 25349383520256.0000 -
mae: 4727422.0000 - val_loss: 24781590102016.0000 - val_mae:
4623896.5000

Epoch 3/100

11/11 _____ 0s 6ms/step - loss: 25349373034496.0000 -
mae: 4727421.0000 - val_loss: 24781577519104.0000 - val_mae:
4623895.0000

Epoch 4/100

11/11 _____ 0s 8ms/step - loss: 25349360451584.0000 -
mae: 4727419.0000 - val_loss: 24781558644736.0000 - val_mae:
4623893.5000

Epoch 5/100

11/11 _____ 0s 6ms/step - loss: 25349335285760.0000 -
mae: 4727416.0000 - val_loss: 24781531381760.0000 - val_mae:
4623890.0000

Epoch 6/100

11/11 _____ 0s 6ms/step - loss: 25349299634176.0000 -
mae: 4727413.0000 - val_loss: 24781483147264.0000 - val_mae:
4623885.5000

Epoch 7/100

11/11 _____ 0s 7ms/step - loss: 25349245108224.0000 -
mae: 4727407.5000 - val_loss: 24781411844096.0000 - val_mae:
4623878.5000

Epoch 8/100

```
11/11 _____ 0s 8ms/step - loss: 25349157027840.0000 -  
mae: 4727398.5000 - val_loss: 24781304889344.0000 - val_mae:  
4623868.5000  
Epoch 9/100  
11/11 _____ 0s 9ms/step - loss: 25349027004416.0000 -  
mae: 4727386.5000 - val_loss: 24781149700096.0000 - val_mae:  
4623853.0000  
Epoch 10/100  
11/11 _____ 0s 10ms/step - loss: 25348838260736.0000 -  
mae: 4727368.5000 - val_loss: 24780929499136.0000 - val_mae:  
4623832.0000  
Epoch 11/100  
11/11 _____ 0s 10ms/step - loss: 25348578213888.0000 -  
mae: 4727344.0000 - val_loss: 24780623314944.0000 - val_mae:  
4623803.5000  
Epoch 12/100  
11/11 _____ 0s 9ms/step - loss: 25348219600896.0000 -  
mae: 4727310.0000 - val_loss: 24780216467456.0000 - val_mae:  
4623766.0000  
Epoch 13/100  
11/11 _____ 0s 9ms/step - loss: 25347735158784.0000 -  
mae: 4727265.5000 - val_loss: 24779677499392.0000 - val_mae:  
4623716.5000  
Epoch 14/100  
11/11 _____ 0s 7ms/step - loss: 25347116498944.0000 -  
mae: 4727209.0000 - val_loss: 24778972856320.0000 - val_mae:  
4623651.5000  
Epoch 15/100  
11/11 _____ 0s 6ms/step - loss: 25346302803968.0000 -  
mae: 4727135.0000 - val_loss: 24778071080960.0000 - val_mae:  
4623569.5000  
Epoch 16/100  
11/11 _____ 0s 6ms/step - loss: 25345279393792.0000 -  
mae: 4727041.0000 - val_loss: 24776934424576.0000 - val_mae:  
4623467.5000  
Epoch 17/100  
11/11 _____ 0s 6ms/step - loss: 25343939313664.0000 -  
mae: 4726922.5000 - val_loss: 24775556595712.0000 - val_mae:  
4623344.0000  
Epoch 18/100  
11/11 _____ 0s 6ms/step - loss: 25342393712640.0000 -  
mae: 4726781.5000 - val_loss: 24773809668096.0000 - val_mae:  
4623188.5000  
Epoch 19/100  
11/11 _____ 0s 6ms/step - loss: 25340399321088.0000 -  
mae: 4726605.0000 - val_loss: 24771750264832.0000 - val_mae:  
4623005.5000  
Epoch 20/100  
11/11 _____ 0s 6ms/step - loss: 25338088259584.0000 -
```

```
mae: 4726396.5000 - val_loss: 24769235779584.0000 - val_mae:
4622783.5000
Epoch 21/100
11/11 _____ 0s 6ms/step - loss: 25335208869888.0000 -
mae: 4726143.5000 - val_loss: 24766310252544.0000 - val_mae:
4622525.0000
Epoch 22/100
11/11 _____ 0s 6ms/step - loss: 25332037976064.0000 -
mae: 4725860.0000 - val_loss: 24762740899840.0000 - val_mae:
4622213.0000
Epoch 23/100
11/11 _____ 0s 6ms/step - loss: 25328078553088.0000 -
mae: 4725507.5000 - val_loss: 24758672424960.0000 - val_mae:
4621857.0000
Epoch 24/100
11/11 _____ 0s 6ms/step - loss: 25323439652864.0000 -
mae: 4725105.0000 - val_loss: 24754023038976.0000 - val_mae:
4621450.0000
Epoch 25/100
11/11 _____ 0s 6ms/step - loss: 25318318407680.0000 -
mae: 4724652.5000 - val_loss: 24748520112128.0000 - val_mae:
4620971.5000
Epoch 26/100
11/11 _____ 0s 6ms/step - loss: 25312123420672.0000 -
mae: 4724111.0000 - val_loss: 24742337708032.0000 - val_mae:
4620434.0000
Epoch 27/100
11/11 _____ 0s 6ms/step - loss: 25305414631424.0000 -
mae: 4723513.0000 - val_loss: 24735133990912.0000 - val_mae:
4619810.0000
Epoch 28/100
11/11 _____ 0s 6ms/step - loss: 25297313333248.0000 -
mae: 4722821.0000 - val_loss: 24727215144960.0000 - val_mae:
4619123.5000
Epoch 29/100
11/11 _____ 0s 6ms/step - loss: 25288461254656.0000 -
mae: 4722046.0000 - val_loss: 24718212071424.0000 - val_mae:
4618345.5000
Epoch 30/100
11/11 _____ 0s 6ms/step - loss: 25278575280128.0000 -
mae: 4721185.5000 - val_loss: 24708015718400.0000 - val_mae:
4617467.0000
Epoch 31/100
11/11 _____ 0s 6ms/step - loss: 25267860930560.0000 -
mae: 4720230.5000 - val_loss: 24696393302016.0000 - val_mae:
4616469.5000
Epoch 32/100
11/11 _____ 0s 6ms/step - loss: 25255171063808.0000 -
mae: 4719142.5000 - val_loss: 24683867013120.0000 - val_mae:
```

```
4615393.5000
Epoch 33/100
11/11 _____ 0s 6ms/step - loss: 25241491341312.0000 -
mae: 4717948.5000 - val_loss: 24670074044416.0000 - val_mae:
4614207.0000
Epoch 34/100
11/11 _____ 0s 6ms/step - loss: 25226452664320.0000 -
mae: 4716639.0000 - val_loss: 24654745960448.0000 - val_mae:
4612891.0000
Epoch 35/100
11/11 _____ 0s 6ms/step - loss: 25209818054656.0000 -
mae: 4715200.5000 - val_loss: 24637870178304.0000 - val_mae:
4611446.0000
Epoch 36/100
11/11 _____ 0s 6ms/step - loss: 25191614775296.0000 -
mae: 4713611.0000 - val_loss: 24619383783424.0000 - val_mae:
4609862.5000
Epoch 37/100
11/11 _____ 0s 6ms/step - loss: 25171431784448.0000 -
mae: 4711889.5000 - val_loss: 24599290970112.0000 - val_mae:
4608144.0000
Epoch 38/100
11/11 _____ 0s 6ms/step - loss: 25150007279616.0000 -
mae: 4709993.5000 - val_loss: 24577159725056.0000 - val_mae:
4606253.0000
Epoch 39/100
11/11 _____ 0s 6ms/step - loss: 25125317509120.0000 -
mae: 4707908.0000 - val_loss: 24553904406528.0000 - val_mae:
4604259.5000
Epoch 40/100
11/11 _____ 0s 6ms/step - loss: 25099786780672.0000 -
mae: 4705677.5000 - val_loss: 24528530964480.0000 - val_mae:
4602088.5000
Epoch 41/100
11/11 _____ 0s 6ms/step - loss: 25072603496448.0000 -
mae: 4703328.5000 - val_loss: 24500387184640.0000 - val_mae:
4599692.5000
Epoch 42/100
11/11 _____ 0s 6ms/step - loss: 25042119294976.0000 -
mae: 4700700.5000 - val_loss: 24470733455360.0000 - val_mae:
4597158.0000
Epoch 43/100
11/11 _____ 0s 6ms/step - loss: 25010643140608.0000 -
mae: 4697944.5000 - val_loss: 24438227599360.0000 - val_mae:
4594385.0000
Epoch 44/100
11/11 _____ 0s 6ms/step - loss: 24975262089216.0000 -
mae: 4694911.5000 - val_loss: 24403897221120.0000 - val_mae:
4591452.0000
```

Epoch 45/100
11/11 _____ 0s 6ms/step - loss: 24938324951040.0000 -
mae: 4691706.5000 - val_loss: 24366727299072.0000 - val_mae:
4588275.0000
Epoch 46/100
11/11 _____ 0s 7ms/step - loss: 24897694728192.0000 -
mae: 4688213.5000 - val_loss: 24327575568384.0000 - val_mae:
4584928.5000
Epoch 47/100
11/11 _____ 0s 6ms/step - loss: 24855730716672.0000 -
mae: 4684564.0000 - val_loss: 24284986605568.0000 - val_mae:
4581292.5000
Epoch 48/100
11/11 _____ 0s 6ms/step - loss: 24809874391040.0000 -
mae: 4680564.0000 - val_loss: 24240168370176.0000 - val_mae:
4577459.0000
Epoch 49/100
11/11 _____ 0s 6ms/step - loss: 24761430179840.0000 -
mae: 4676379.5000 - val_loss: 24192760152064.0000 - val_mae:
4573396.5000
Epoch 50/100
11/11 _____ 0s 6ms/step - loss: 24710637158400.0000 -
mae: 4672007.0000 - val_loss: 24141358956544.0000 - val_mae:
4569003.0000
Epoch 51/100
11/11 _____ 0s 6ms/step - loss: 24655500935168.0000 -
mae: 4667202.0000 - val_loss: 24087478927360.0000 - val_mae:
4564385.0000
Epoch 52/100
11/11 _____ 0s 6ms/step - loss: 24598244491264.0000 -
mae: 4662149.0000 - val_loss: 24029832413184.0000 - val_mae:
4559448.0000
Epoch 53/100
11/11 _____ 0s 6ms/step - loss: 24535919230976.0000 -
mae: 4656790.0000 - val_loss: 23970071969792.0000 - val_mae:
4554313.0000
Epoch 54/100
11/11 _____ 0s 6ms/step - loss: 24471192731648.0000 -
mae: 4651120.0000 - val_loss: 23906865905664.0000 - val_mae:
4548877.5000
Epoch 55/100
11/11 _____ 0s 6ms/step - loss: 2440422279680.0000 -
mae: 4645287.5000 - val_loss: 23838953832448.0000 - val_mae:
4543048.5000
Epoch 56/100
11/11 _____ 0s 6ms/step - loss: 24330614341632.0000 -
mae: 4638929.0000 - val_loss: 23769007521792.0000 - val_mae:
4537003.5000
Epoch 57/100

```
11/11 _____ 0s 6ms/step - loss: 24255672614912.0000 -  
mae: 4632313.5000 - val_loss: 23694443282432.0000 - val_mae:  
4530580.5000  
Epoch 58/100  
11/11 _____ 0s 6ms/step - loss: 24177809555456.0000 -  
mae: 4625353.0000 - val_loss: 23614996873216.0000 - val_mae:  
4523723.0000  
Epoch 59/100  
11/11 _____ 0s 6ms/step - loss: 24092512092160.0000 -  
mae: 4617948.5000 - val_loss: 23533761593344.0000 - val_mae:  
4516694.0000  
Epoch 60/100  
11/11 _____ 0s 6ms/step - loss: 24004102455296.0000 -  
mae: 4610246.5000 - val_loss: 23449581912064.0000 - val_mae:  
4509377.5000  
Epoch 61/100  
11/11 _____ 0s 6ms/step - loss: 23913369174016.0000 -  
mae: 4602258.5000 - val_loss: 23360352288768.0000 - val_mae:  
4501617.5000  
Epoch 62/100  
11/11 _____ 0s 6ms/step - loss: 23820385648640.0000 -  
mae: 4593875.0000 - val_loss: 23264720060416.0000 - val_mae:  
4493298.5000  
Epoch 63/100  
11/11 _____ 0s 6ms/step - loss: 23715427385344.0000 -  
mae: 4584807.5000 - val_loss: 23170436300800.0000 - val_mae:  
4485029.0000  
Epoch 64/100  
11/11 _____ 0s 6ms/step - loss: 23616519405568.0000 -  
mae: 4575914.0000 - val_loss: 23067675852800.0000 - val_mae:  
4476032.5000  
Epoch 65/100  
11/11 _____ 0s 6ms/step - loss: 23507389906944.0000 -  
mae: 4566115.5000 - val_loss: 22962067472384.0000 - val_mae:  
4466751.5000  
Epoch 66/100  
11/11 _____ 0s 6ms/step - loss: 23392946225152.0000 -  
mae: 4556050.5000 - val_loss: 22854120767488.0000 - val_mae:  
4457222.0000  
Epoch 67/100  
11/11 _____ 0s 6ms/step - loss: 23277019856896.0000 -  
mae: 4545692.5000 - val_loss: 22740497072128.0000 - val_mae:  
4447181.5000  
Epoch 68/100  
11/11 _____ 0s 6ms/step - loss: 23154267258880.0000 -  
mae: 4534626.0000 - val_loss: 22623381618688.0000 - val_mae:  
4436783.5000  
Epoch 69/100  
11/11 _____ 0s 6ms/step - loss: 23030338158592.0000 -
```



```
mae: 4523301.0000 - val_loss: 22499936960512.0000 - val_mae:
4425792.0000
Epoch 70/100
11/11 _____ 0s 6ms/step - loss: 22898735579136.0000 -
mae: 4511481.5000 - val_loss: 22372688068608.0000 - val_mae:
4414427.0000
Epoch 71/100
11/11 _____ 0s 7ms/step - loss: 22761871245312.0000 -
mae: 4499072.5000 - val_loss: 22242767405056.0000 - val_mae:
4402769.5000
Epoch 72/100
11/11 _____ 0s 7ms/step - loss: 22625132740608.0000 -
mae: 4486522.0000 - val_loss: 22104021925888.0000 - val_mae:
4390313.5000
Epoch 73/100
11/11 _____ 0s 6ms/step - loss: 22480303423488.0000 -
mae: 4473123.0000 - val_loss: 21962726309888.0000 - val_mae:
4377561.0000
Epoch 74/100
11/11 _____ 0s 6ms/step - loss: 22331072184320.0000 -
mae: 4459316.5000 - val_loss: 21818043793408.0000 - val_mae:
4364424.5000
Epoch 75/100
11/11 _____ 0s 6ms/step - loss: 22173775298560.0000 -
mae: 4445033.0000 - val_loss: 21672251883520.0000 - val_mae:
4351104.5000
Epoch 76/100
11/11 _____ 0s 6ms/step - loss: 22018252603392.0000 -
mae: 4430496.0000 - val_loss: 21518906032128.0000 - val_mae:
4337043.5000
Epoch 77/100
11/11 _____ 0s 6ms/step - loss: 21855366807552.0000 -
mae: 4415407.5000 - val_loss: 21360671719424.0000 - val_mae:
4322475.5000
Epoch 78/100
11/11 _____ 0s 6ms/step - loss: 21686850158592.0000 -
mae: 4399505.5000 - val_loss: 21197920141312.0000 - val_mae:
4307398.5000
Epoch 79/100
11/11 _____ 0s 6ms/step - loss: 21515189878784.0000 -
mae: 4383220.5000 - val_loss: 21029952946176.0000 - val_mae:
4291776.5000
Epoch 80/100
11/11 _____ 0s 6ms/step - loss: 21334943858688.0000 -
mae: 4366314.0000 - val_loss: 20860112994304.0000 - val_mae:
4275866.0000
Epoch 81/100
11/11 _____ 0s 6ms/step - loss: 21152898482176.0000 -
mae: 4348987.0000 - val_loss: 20684090638336.0000 - val_mae:
```

```
4259299.5000
Epoch 82/100
11/11 _____ 0s 6ms/step - loss: 20966667190272.0000 -
mae: 4331244.0000 - val_loss: 20503504879616.0000 - val_mae:
4242216.0000
Epoch 83/100
11/11 _____ 0s 6ms/step - loss: 20771223109632.0000 -
mae: 4312484.0000 - val_loss: 20321847476224.0000 - val_mae:
4224867.5000
Epoch 84/100
11/11 _____ 0s 6ms/step - loss: 20577920221184.0000 -
mae: 4293529.5000 - val_loss: 20132954898432.0000 - val_mae:
4206718.5000
Epoch 85/100
11/11 _____ 0s 6ms/step - loss: 20381322706944.0000 -
mae: 4274355.5000 - val_loss: 19936030228480.0000 - val_mae:
4187735.2500
Epoch 86/100
11/11 _____ 0s 6ms/step - loss: 20169992699904.0000 -
mae: 4253668.0000 - val_loss: 19740957343744.0000 - val_mae:
4168735.2500
Epoch 87/100
11/11 _____ 0s 6ms/step - loss: 19964373237760.0000 -
mae: 4233005.5000 - val_loss: 19537665720320.0000 - val_mae:
4148806.5000
Epoch 88/100
11/11 _____ 0s 6ms/step - loss: 19753643016192.0000 -
mae: 4212118.0000 - val_loss: 19328936181760.0000 - val_mae:
4128225.5000
Epoch 89/100
11/11 _____ 0s 6ms/step - loss: 19527928643584.0000 -
mae: 4189461.0000 - val_loss: 19124283506688.0000 - val_mae:
4107799.2500
Epoch 90/100
11/11 _____ 0s 6ms/step - loss: 19308054839296.0000 -
mae: 4167502.7500 - val_loss: 18915270852608.0000 - val_mae:
4086792.7500
Epoch 91/100
11/11 _____ 0s 6ms/step - loss: 19085687521280.0000 -
mae: 4144410.5000 - val_loss: 18699440357376.0000 - val_mae:
4064898.2500
Epoch 92/100
11/11 _____ 0s 6ms/step - loss: 18858882629632.0000 -
mae: 4121067.0000 - val_loss: 18477182091264.0000 - val_mae:
4042185.7500
Epoch 93/100
11/11 _____ 0s 6ms/step - loss: 18621361291264.0000 -
mae: 4096403.7500 - val_loss: 18257788534784.0000 - val_mae:
4019500.0000
```

```
Epoch 94/100
11/11 _____ 0s 6ms/step - loss: 18391467294720.0000 -
mae: 4072027.5000 - val_loss: 18029589037056.0000 - val_mae:
3995768.0000
Epoch 95/100
11/11 _____ 0s 6ms/step - loss: 18148426252288.0000 -
mae: 4046525.7500 - val_loss: 17802899488768.0000 - val_mae:
3971894.5000
Epoch 96/100
11/11 _____ 0s 6ms/step - loss: 17910317711360.0000 -
mae: 4020785.7500 - val_loss: 17570111422464.0000 - val_mae:
3947147.2500
Epoch 97/100
11/11 _____ 0s 6ms/step - loss: 17663994626048.0000 -
mae: 3994228.5000 - val_loss: 17337507905536.0000 - val_mae:
3922125.7500
Epoch 98/100
11/11 _____ 0s 6ms/step - loss: 17414429343744.0000 -
mae: 3967224.2500 - val_loss: 17105164435456.0000 - val_mae:
3896890.2500
Epoch 99/100
11/11 _____ 0s 6ms/step - loss: 17168296050688.0000 -
mae: 3939829.7500 - val_loss: 16866470789120.0000 - val_mae:
3870657.5000
Epoch 100/100
11/11 _____ 0s 6ms/step - loss: 16920006885376.0000 -
mae: 3912213.2500 - val_loss: 16622188232704.0000 - val_mae:
3843539.7500
```

```
loss, mae = model.evaluate(X_test, y_test)
```

```
print("Test Loss (MSE):", loss)
```

```
print("Test MAE:", mae)
```

```
4/4 _____ 0s 5ms/step - loss: 20055169433600.0000 -
```

```
mae: 4147622.7500
```

```
Test Loss (MSE): 20055169433600.0
```

```
Test MAE: 4147622.75
```

```
y_pred = model.predict(X_test)
```

```
print(y_pred[:10])
```

```
4/4 _____ 0s 13ms/step
```

```
[[2.5585541e+05]
```

```
 [2.0040624e+06]
```

```
 [3.3709528e+05]
```

```
 [3.5324262e+05]
```

```
 [2.5565256e+05]
```

```
 [2.8822250e+05]
```

```
[1.3431506e+06]
[2.1173990e+06]
[2.2408141e+05]
[2.8779484e+05]]
```

```
from sklearn.metrics import mean_squared_error, r2_score
import numpy as np
```

```
rmse = np.sqrt(mean_squared_error(y_test, y_pred))
r2 = r2_score(y_test, y_pred)
```

```
print("RMSE :", rmse)
print("R2 Score :", r2)
```

```
RMSE : 4478299.837393651
R2 Score : -2.9677295684814453
```

```
comparison = pd.DataFrame({
    "Actual Price": y_test.values,
    "Predicted Price": y_pred.flatten()
})
```

```
comparison.head(10)
```

	Actual Price	Predicted Price
0	4060000	2.558554e+05
1	6650000	2.004062e+06
2	3710000	3.370953e+05
3	6440000	3.532426e+05
4	2800000	2.556526e+05
5	4900000	2.882225e+05
6	5250000	1.343151e+06
7	4543000	2.117399e+06
8	2450000	2.240814e+05
9	3353000	2.877948e+05

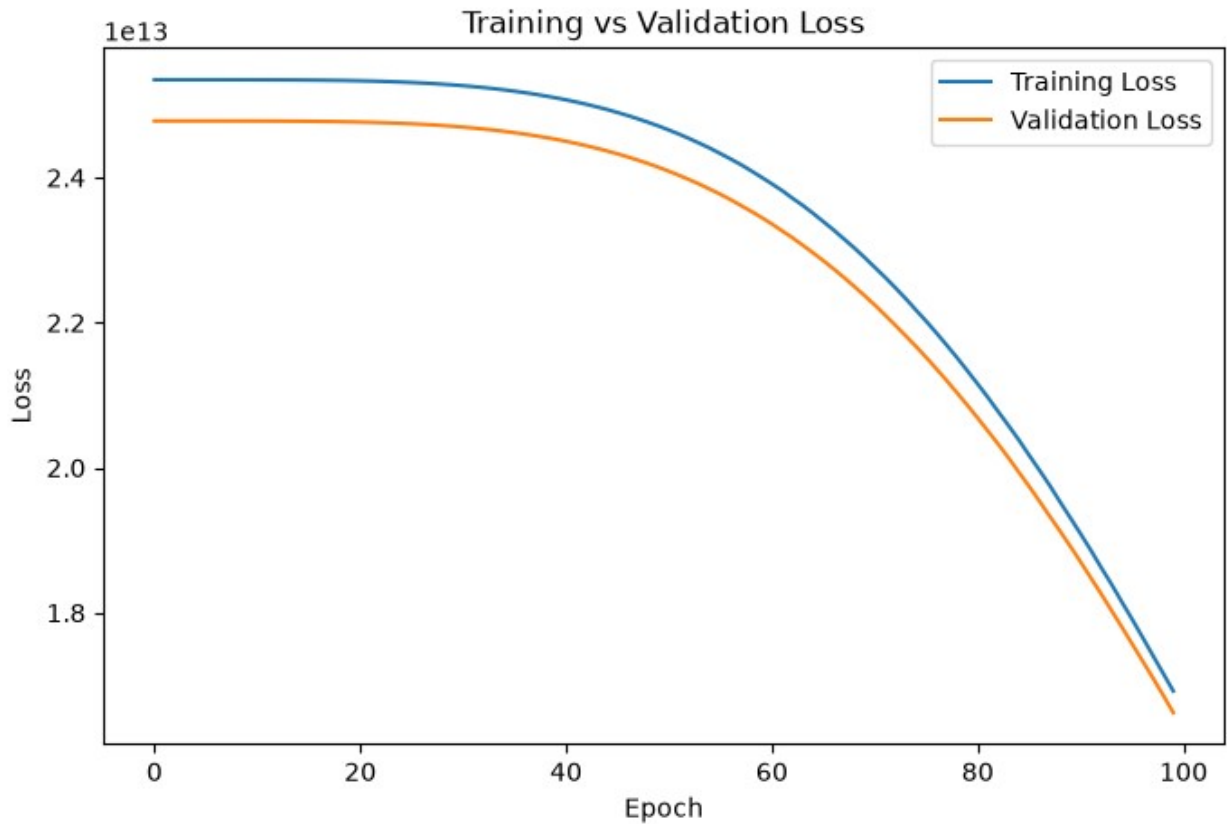
```
import matplotlib.pyplot as plt
```

```
plt.figure(figsize=(8,5))
```

```
plt.plot(history.history['loss'], label='Training Loss')
plt.plot(history.history['val_loss'], label='Validation Loss')
```

```
plt.title("Training vs Validation Loss")
plt.xlabel("Epoch")
plt.ylabel("Loss")
plt.legend()
```

```
plt.show()
```

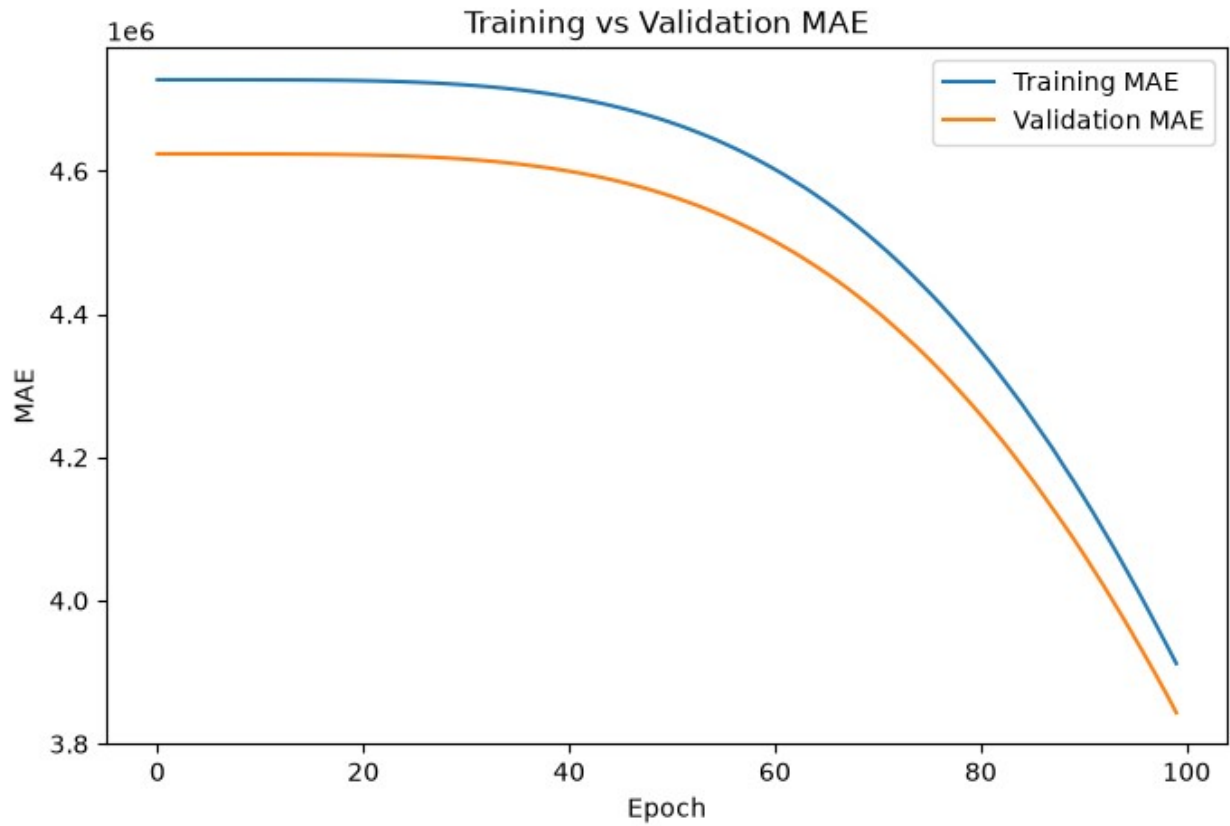


```
plt.figure(figsize=(8,5))

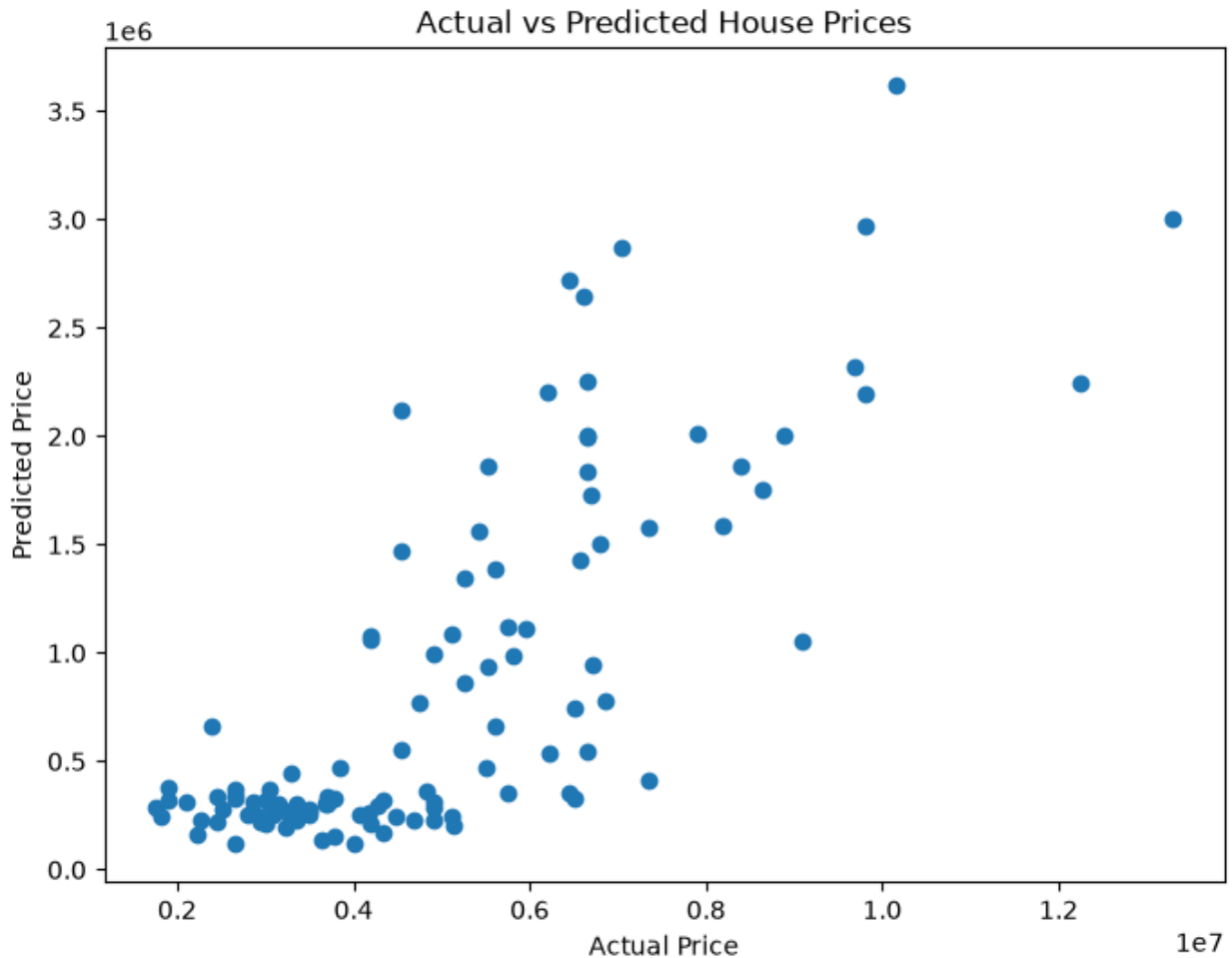
plt.plot(history.history['mae'], label='Training MAE')
plt.plot(history.history['val_mae'], label='Validation MAE')

plt.title("Training vs Validation MAE")
plt.xlabel("Epoch")
plt.ylabel("MAE")
plt.legend()

plt.show()
```



```
plt.figure(figsize=(8,6))  
plt.scatter(y_test, y_pred)  
plt.xlabel("Actual Price")  
plt.ylabel("Predicted Price")  
plt.title("Actual vs Predicted House Prices")  
plt.show()
```



```
sample = np.array([[7420,4,2,3,1,0,1,0,1,2,1,2]])
```

```
sample = scaler.transform(sample)
```

```
prediction = model.predict(sample)
```

```
print("Predicted House Price :", prediction[0][0])
```

```
1/1 _____ 0s 28ms/step
```

```
Predicted House Price : 2.7167695e+06
```

```
d:\B-Tech\7th Sem_BTech\ADS_Lab\Assignment_01\tfenv\Lib\site-packages\
sklearn\utils\validation.py:2827: UserWarning: X does not have valid
feature names, but StandardScaler was fitted with feature names
  warnings.warn(
```

```
model.save("house_price_ann_model.keras")
```

```
print("Model Saved Successfully!")
```

```
Model Saved Successfully!
```

```
from tensorflow.keras.models import load_model
loaded_model = load_model("house_price_ann_model.keras")
prediction = loaded_model.predict(sample)
print(prediction)
1/1 _____ 0s 52ms/step
[[2.7167695e+06]]
```